Orthogonal tori and their projective orders

Human Compression project

This working note supplies the torus interface for the orthogonal part of the generation argument. It is not yet a reviewed component. It does not classify maximal overgroups or estimate the centralizers of reversing involutions. In particular it does not claim the full existence theorem.

Put $E = \mathbb{F}_{q^{2k}}$ and $E_0 = \mathbb{F}_{q^k}$, and give E, as a vector space over $\mathbb{F}_q$, the quadratic form

$$Q(v) = \mathrm{Tr}_{E_0/\mathbb{F}_q}(cv^{q^k+1}), \qquad c \in E_0^\times.$$

It is a nondegenerate form of minus type. Here is also a direct check of the type: every nonzero value of the norm has q^k+1 preimages, so the number of zeros is $1 + (q^{k-1}-1)(q^k+1) = q^{2k-1} - (q-1)q^{k-1}$. Nondegeneracy follows from the nondegenerate trace pairing and the polar form of the quadratic norm. Multiplication by the norm-one group is a cyclic group of isometries of order q^k+1.

Lemma 1. *Suppose q is odd and z generates this norm-one group, with $q^k+1 > 2$. Multiplication by z has nontrivial spinor norm.*

Proof. We use the determinant formula for the spinor norm in [1, Definition 3.1]; this formula and its homomorphism property are borrowed. Since -1 is not an eigenvalue of multiplication by z, that formula is the square class of $\det((1+z)/2) = \mathrm{N}_{E/\mathbb{F}_q}(1+z)/2^{2k}$. Choose $w \in E$ of order $2(q^k+1)$ with $w^2 = z$. Such a choice exists because $2(q^k+1)$ divides $q^{2k}-1$. Then $w^{q^k} = -w^{-1}$. For $u = w + w^{-1}$ we have $u \neq 0$ and $u^{q^k} = -u$, and hence $u^2 \in E_0$ is a nonsquare there. Indeed a square root in E_0 would have to be u or $-u$. Now

$$\mathrm{N}_{E/E_0}(1+z) = (1+z)(1+z^{-1}) = u^2.$$

The norm from a finite extension of finite fields sends a generator of the multiplicative group to a generator. Consequently it sends a nonsquare to a nonsquare. The displayed determinant has nonsquare class, as claimed. Multiplication by z has determinant one, so its powers lying in Ω are precisely its even powers. □

Lemma 2. *Let $q \geq 2$, $k \geq 1$, and $M > 2$ divide q^k+1. If $M > q^{k/3}+1$, then the multiplicative order of q modulo M is $2k$.*

Proof. That order is $2d$ with $d \mid k$ and k/d odd: it divides $2k$, but the power q^k has order two. If $d < k$, then $d \leq k/3$, and $q^d \equiv q^k \equiv -1 \pmod{M}$. Thus $M \leq q^d+1$, contrary to the assumption. □

The required inequalities hold throughout the ranges below. For odd $q \geq 3$ and $k \geq 3$, $(q^k+1)/4 > q^{k/3}+1$: setting $y = q^{k/3} \geq 3$, this is $y^3 - 4y - 3 > 0$, which follows from its value and positive derivative at $y = 3$. For even q we use $M = q^k+1$ directly, with $k \geq 3$.

Proposition 3. *Use the following natural orthogonal isometries and let x denote their projective images.*

1. *In minus dimension $2m$, $m \geq 4$, put $N = q^m + 1$. Use a primitive norm-one multiplier g when q is even and g^2 when q is odd. Its projective order is N in even characteristic and $N/\gcd(4, N)$ in odd characteristic.*

2. *In odd dimension $2m + 1$, q odd and $m \geq 3$, use g^2 on a negative $2m$-space and the identity on its orthogonal line. Its order is $(q^m + 1)/2$.*

3. *In plus dimension $2m$, $m \geq 4$, put $A = q^{m-1} + 1$ and $B = q + 1$. On negative spaces of dimensions $2m - 2$ and 2, use primitive norm-one multipliers g, h. If m is even, choose the identification of the two norm tori so that $h = g^{A/B}$. For even q the projective order is AB when m is odd and A when m is even. For odd q it is*

$$\begin{cases} AB/\gcd(4, q^m - 1), & m \text{ odd}, \\ A/2, & m \text{ even}. \end{cases}$$

All these isometries belong to the indicated omega groups.

Proof. In odd characteristic, Lemma 1 gives omega membership in the first two cases and cancellation of the two nontrivial spinor norms in the third. In even characteristic the elements have odd order, so their images under the Dickson homomorphism to a group of order two vanish. These homomorphisms and the determinant condition defining omega are as in [1, Section 3].

The order of g^2 in the first case is $N/2$. The scalar $-I$ belongs to this cyclic subgroup if and only if $4 \mid N$, which gives exactly the additional divisor stated. In odd dimension the fixed line prevents any power other than the identity from being scalar.

For the third case, if m is odd, the Euclidean algorithm gives $\gcd(A, B) = \gcd(2, q+1)$. For odd q, A has exactly one factor of two. The pair (g, h) has order $AB/2$; its half-order power is $(-I, -I)$ exactly when B also has exactly one factor of two, namely when $q \equiv 1 \pmod 4$. If m is even, $B \mid A$ and A/B is odd. The pair has order A, and for odd q its $A/2$ power is $(-I, -I)$. There is no nontrivial scalar isometry in even characteristic. These observations prove all the projective order formulas. □

Proposition 4. *In the natural projective conformal orthogonal group, the centralizers of the elements in Proposition 3 are tori of orders N, N, AB, respectively. They lie in the connected adjoint component. There is no involutory reversing automorphism represented on this natural module by a semilinear map with nontrivial ordinary field part. The assertion does not address the additional triality automorphisms in plus dimension eight.*

Proof. On the large negative space let α be the chosen multiplier, and write M for its order after identifying scalar signs. In the minus case $M = N/\gcd(4, N)$ for odd q; in the plus case it is $A/2$ for odd q; in even characteristic it is the full norm-one order. In the odd dimensional case we may use the smaller value $N/\gcd(4, N)$ too. Lemma 2 and the inequalities following it show that $\operatorname{ord}_M(q) = 2k$, where $k = m$ or $m - 1$. In particular the large block is irreducible, and is distinguished from the other blocks by its dimension.

If a conformal map Y centralizes the projective image of the chosen isometry X, then $YXY^{-1} = \lambda X$. Comparing the preserved forms gives $\lambda^2 = 1$. On the large block the eigenvalues of the two similar matrices therefore give $\alpha^{q^j} = \lambda\alpha$ for some $0 \leq j < 2k$. Modulo scalar signs this says $q^j \equiv 1 \pmod M$, so $j = 0$ and $\lambda = 1$. Hence Y centralizes X as a linear map.

Its restriction to the large block belongs to the field algebra $E = \mathbb{F}_q[X]$, and the conformal condition on multiplication by z is $\mathrm{N}_{E/E_0}(z) \in \mathbb{F}_q^\times$. This follows by substituting into Q and using the nondegenerate trace pairing. There are $(q-1)(q^k+1)$ such multipliers. In minus type division by the $q - 1$ scalar multipliers gives N. In plus type the two block multipliers must have the same norm in $\mathbb{F}_q^\times$. The number of pairs is $(q - 1)AB$, so their projective group has order AB. The small block is irreducible because its primitive multiplier has order $q+1>2$; this includes $q=2$. In odd dimension the line multiplier a forces the large-block norm to be a^2. Dividing both multipliers by a leaves precisely the norm-one group, of order N. These multiplicative centralizers are connected tori over the algebraic closure and consequently belong to the connected adjoint component.

Finally let $q=p^f$. An involutory automorphism with a nontrivial field projection has f even and projection $p^{f/2}$. If it reverses x, comparison of the eigenvalues on the large block, again allowing the scalar sign, implies

$$p^{f/2+fj} \equiv -1 \pmod{M}$$

for some j (the sign introduced by a scalar is already quotiented out in M). Squaring gives $q^{2j+1} \equiv 1 \pmod{M}$. This contradicts $\operatorname{ord}_M(q)=2k$. Natural graph automorphisms are represented by linear conformal maps and do not alter this argument; exceptional triality still requires a separate argument. ☐

The distinction between a cyclic subgroup and its centralizer matters. For example in plus type with even m, x has order $A/2$ for odd q but its adjoint torus has order AB. In minus dimension ten over $\mathbb{F}_3$, the projective Singer image has order 61, whereas the natural omega Singer subgroup has order 122. The latter convention agrees with [2, Table 1]. We retain these scalar losses explicitly in the later counting argument.

References

[1] S. H. Murray and C. M. Roney-Dougal, *The spinor norm and homomorphism algorithms for classical groups*, preprint dated 2 May 2008, Definition 3.1 and Lemma 3.2.

[2] D. Bubboloni, M. S. Lucido and T. Weigel, *2-Coverings of classical groups*, arXiv:1102.0660 (2011).